

PHAM TIEN DUNG

Data Analyst | Data Scientist | Quantitative Finance

Mathematical Economics · Machine Learning · Financial Analytics

CONTACT

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PROFESSIONAL SUMMARY

Final-year Mathematical Economics student at the National Economics University (NEU), Vietnam (GPA: 3.2/4.0), with a rigorous foundation in quantitative methods, statistical modeling, and Python-based data analysis. Demonstrated ability to design end-to-end machine learning pipelines — from feature engineering through model evaluation — applied to real-world financial problems including credit risk scoring and portfolio optimization. Founder and President of CMET (30+ members), a club bridging mathematics, technology, and quantitative finance. Seeking a Data Scientist internship to deliver data-driven insights and predictive models that support evidence-based decisions, with a long-term goal of specializing in Financial Data Science.

EDUCATION

National Economics University (NEU) — B.Sc. Mathematical Economics

2023 – Present

Faculty of Mathematical Economics | GPA: 3.2 / 4.0 | Hanoi, Vietnam

- Core coursework: Probability & Mathematical Statistics, Econometrics, Optimization Theory (LP/QP/NLP), Time Series Analysis, Financial Mathematics, Portfolio Theory, Stochastic Processes, Linear Algebra, Real Analysis
- Awards: Third Prize — National Student Mathematics Olympiad (Algebra, 2025); Top 5 — Calculus Olympiad & Top 8 — Algebra Olympiad, University Level (2026)

PROJECTS & EXPERIENCE

Hybrid Credit Scoring Model for Gen Z Borrowers

2025

Independent Research | Machine Learning · Credit Risk · Python

- Designed a hybrid credit scoring system combining traditional financial features (income, debt-to-income ratio, repayment history) with digital behavioral signals (app usage patterns, transaction frequency) targeting the underserved Gen Z lending segment.
- Algorithms & Stack:** Logistic Regression, Random Forest, XGBoost (Scikit-learn); feature engineering with Pandas/NumPy; hyperparameter tuning via GridSearchCV; SHAP values for model explainability and regulatory interpretability.
- Aligned model architecture with Basel II/III regulatory standards — estimated Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD) under the Internal Ratings-Based (IRB) approach.
- Result: AUC > 0.80 on holdout test set; SHAP analysis confirmed digital behavioral features contribute ~35% of predictive power, validating the hybrid data strategy over traditional-only models.

Multi-Asset Portfolio Optimization

2025

Mathematical Economic Modeling Competition — NEU | Optimization · Monte Carlo · Python

- Formulated a Mean-Variance (Markowitz) optimization problem across four asset classes in a simulated market: bank deposits, precious metals, cryptocurrency, and equities with varied risk-return profiles.
- Implemented Monte Carlo simulation (10,000 scenarios) to estimate portfolio-level VaR (95%, 99%) and CVaR; plotted the efficient frontier and identified the Maximum Sharpe Ratio portfolio using SciPy constrained optimization.
- Integrated binding risk constraints (VaR budget ceiling, sector concentration caps) and conducted sensitivity analysis on expected return assumptions to surface robust allocations across stress regimes.
- Delivered a ranked portfolio strategy with risk-adjusted return comparison and stress-tested drawdown scenarios; work presented at NEU competition.

FX Exchange Rate Risk Analysis & Stress Testing

2026

Financial Risk Management Coursework — NEU | ARIMA · GARCH · VaR · Stress Testing

- ▶ Applied Augmented Dickey-Fuller (ADF) test to confirm stationarity of the VND/USD daily exchange rate series; built ARIMA and GARCH(1,1) models in Python (Statsmodels) for volatility forecasting and conditional variance estimation.
- ▶ Estimated parametric VaR at 99% confidence (10-day holding period) under normal and fat-tailed distribution assumptions; benchmarked against historical simulation VaR to assess model risk.
- ▶ Designed multi-scenario stress tests (5–15% abrupt FX shock) to quantify P&L impact, capital adequacy ratio (CAR) deterioration, and net interest margin compression; proposed FX hedging strategies via forward contracts and vanilla options.
- ▶ Analyzed ROA and ROE sensitivity of a representative commercial bank under adverse FX scenarios — translating quantitative model outputs into business-readable risk narratives.

Founder & President — CMET - Club of Mathematical Economics & Technology

2023 – Present

Club of Mathematical Economics & Technology | National Economics University | 30+ members

- ▶ Founded and scaled a 30+ member community bridging mathematics, data science, and quantitative finance at NEU; established CMET as a recognized academic club with multi-stakeholder reach (students, faculty, industry).
- ▶ Organized 3 flagship talks: "Personal Data Security in the Digital Age" (Apr 2026, 200+ attendees), "Quantitative Tools — From Lecture Hall to Practice" (Oct 2025), and "Study Abroad & International Career Opportunities in Math, Data & Economics" (Apr 2024).
- ▶ Designed and delivered hands-on workshops on Python for Finance, statistical modeling, and data visualization; mentored members on research methodology, competition strategy, and technical project development.
- ▶ Building a sustainable ecosystem connecting students with industry professionals and faculty through talkshow series, peer-learning sessions, and enterprise partnerships.

TECHNICAL SKILLS

Programming	Python (NumPy, Pandas, Scikit-learn, Statsmodels, Matplotlib, Seaborn, SciPy) SQL (querying, joins, aggregation) Excel / VBA (Financial Modeling)
Machine Learning	Supervised Learning (Logistic Regression, Decision Trees, Random Forest, XGBoost); Model Evaluation (AUC-ROC, KS Statistic, Gini Coefficient); Hyperparameter Tuning (GridSearchCV); SHAP Explainability; Cross-validation
Statistical Methods	Probability Theory, Regression Analysis (OLS / WLS), Hypothesis Testing, Time Series Analysis (ARIMA / GARCH), Monte Carlo Simulation, Cluster Analysis, A/B Testing (fundamentals)
Risk & Finance	Value at Risk (VaR / CVaR), Portfolio Optimization (Mean-Variance / Markowitz), Stress Testing & Scenario Analysis, Basel II/III Framework, Credit Risk Modeling (PD / LGD / EAD), Financial Statement Analysis (ROA / ROE)
Visualization	Matplotlib, Seaborn, Plotly Excel Charts Power BI (basic)
Languages	Vietnamese — Native English — Professional (academic reading & writing, technical documentation)

ACHIEVEMENTS & ACTIVITIES

- ▶ Mathematics Olympiad (2024 & 2026): Third Prize — National Student Olympiad (Algebra, 2025); Top 5 Calculus & Top 8 Algebra, University Level (2026) — evidence of sustained, high-level quantitative problem-solving ability.
- ▶ Talkshow Organizing Committee Lead: Curated and executed CMET Technology Talkshow Series at NEU covering AI & LLMs, Big Data in Finance, Personal Data Security, and Quantitative Tools in Practice.
- ▶ Continuing Education: CFA Level 1 — currently studying (CFA Institute Curriculum); FRM Part 1 — self-study (GARP Study Materials).
- ▶ Reference texts: Risk Management & Financial Institutions (Hull); Python for Data Analysis (McKinney); The Elements of Statistical Learning (Hastie et al.); Advances in Financial Machine Learning (de Prado).
- ▶ Competitive Athletics: Former provincial and national-level track & field athlete (sprint / relay) — instilled high-performance discipline, pressure tolerance, and systematic goal-setting directly applicable to data project execution.